

Architecture of the MDNRef Model

Bridging Analytical Physics and Machine Learning for DSP-FEC
Channel Interfaces in DCI Modems

DSP-FEC Interface • 3djax Project • System Specification

The Fundamental Tension

Physics-Based Rigid Models

Hardcoding deterministic formulas (like standard COM) assumes perfect knowledge of DSP behavior. While grounded, they fail to capture the stochastic, non-linear realities of hardware implementation.

Unconstrained Neural Networks

Purely data-driven models lose physical grounding. Without a "Rosetta Stone" to relate parameters back to S-parameters or fiber physics, the model becomes a black box unusable by analog teams.

*Solution: Use the **Impairment Space** as the unifying coordinate system.*

The 5-D Impairment Space



Signal & ISI

Asig & Aisi: Capture the available main cursor and residual tails from tau_fall or reflections.



Crosstalk

Axtalk: Integrated Crosstalk Noise (ICN) derived from neighbor lane isolation and aggressor count.



Dynamic

Adyn: Tracking penalties caused by thermal drift and mechanical vibration (SOP/Connector).



Noise

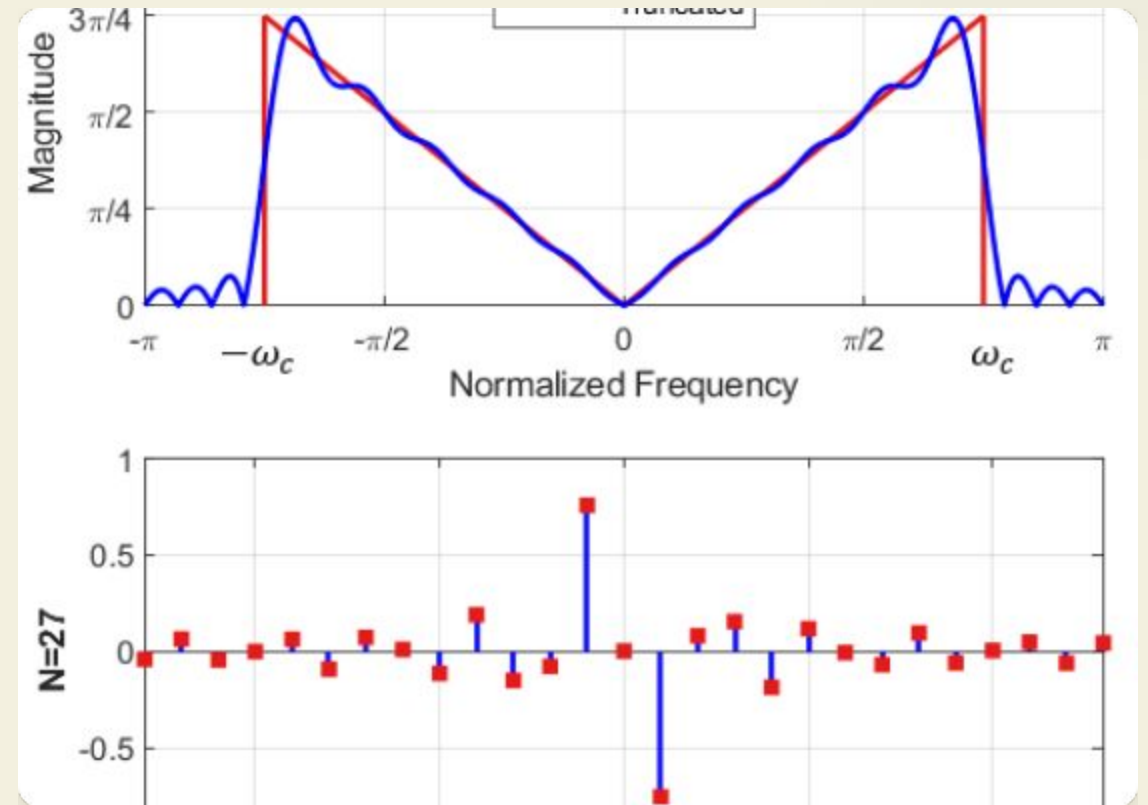
Anoise: Baseline thermal noise floor scaled by peak-to-RMS factors for bit-error targets.

The S-Parameter Unifier

Projecting $h(t)$ to Vectors

Whether derived from double-exponential analytic models or measured **S21 Touchstone files**, every channel is just an array of taps.

- ✓ **Universal Projection:** Map any impulse response onto the 5 impairment axes.
- ✓ **Aisi Calculation:** Sum of un-equalized tails regardless of physical origin (reflection vs. smearing).



Virtual SNR: The Objective Referee

Evolving COM from a rigid formula to a baseline **Virtual SNR** computed via MMSE of a theoretical infinite-length equalizer.

$$\text{COM}_{\text{virtual}} = 10\log_{10} \frac{1 - \sigma_e^2}{\sigma_e^2}$$



Calculate MMSE: Derive theoretical minimum error variance from channel $h(t)$ and noise. σ_e^2



Physics-Based Metric: Provides an objective "referee" score for YAML profiles and future S-parameter ingestion.



Unified View: Every channel becomes a single point in the 5-D impairment space.

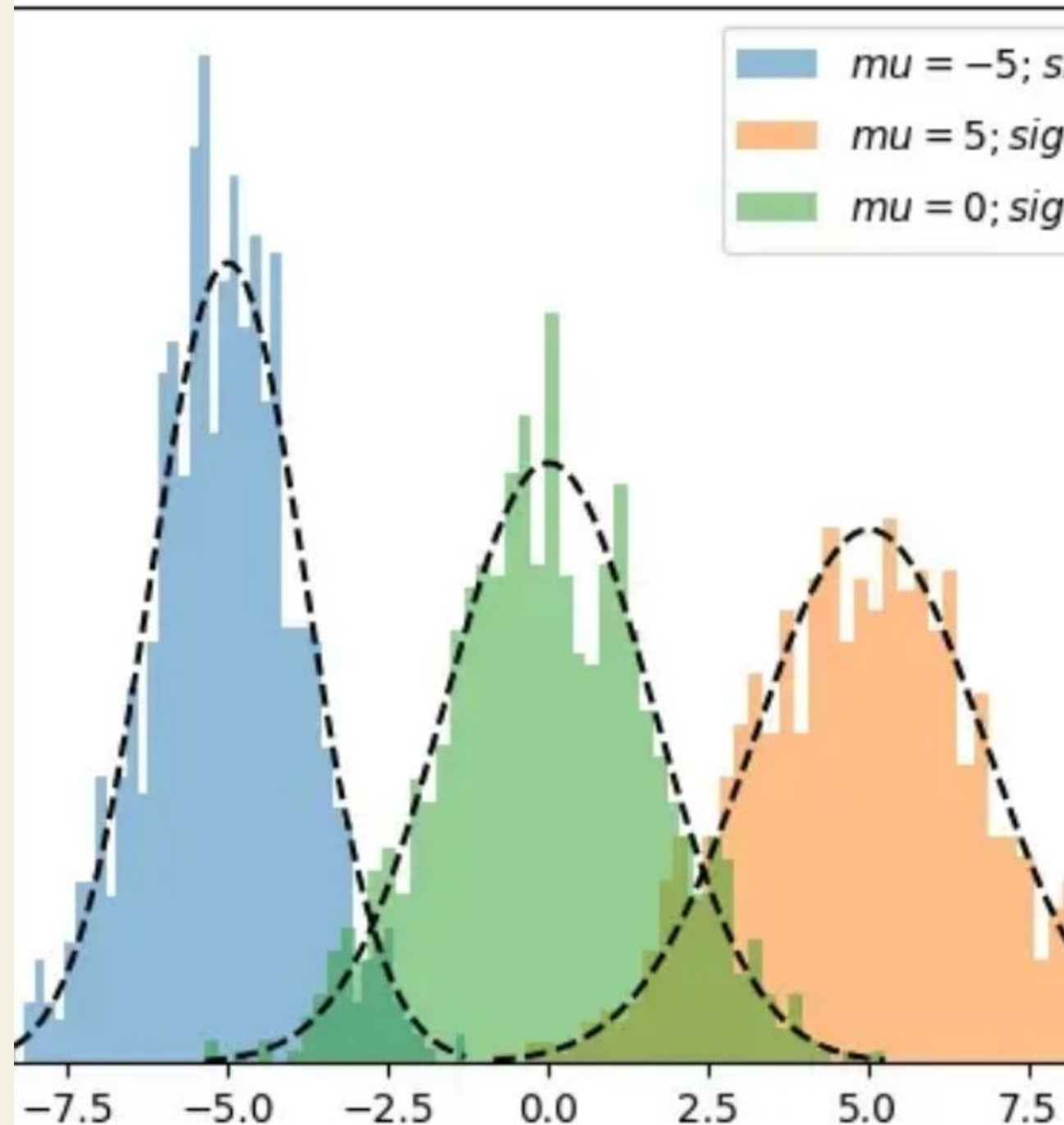
The Role of the MDN

Predicting the Post-DSP Reality

The MDN does not predict COM; it **consumes** the impairment vector to predict the actual LLR distributions the FEC will receive.

It learns the *delta* between theoretical MMSE limits and real DSP implementation (finite taps, PI loop lag, tracking slips).

When tracking error (Adyn) crosses frequency thresholds, MDN captures the non-Gaussian "tails" in LLR distributions.



Structuring Stress Classes (A-E)

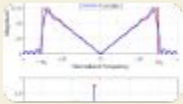
Class	Focus Area	Impairment Vector Dominance
A (Bring-up)	Ideal Pass-thru	Pure AWGN (Anoise), Zero ISI/Dyn.
B (Loss-Dom)	Noise Enhancement	Deep Anoise, Minimal Reflections.
C (Reflection)	LMS/PI Tracking	High Aisi and Adyn (Vibration/Drift).
D (Crosstalk)	Non-Gaussian Interference	High Axtalk (Aggr. Lane Isolation).
E (Experimental)	Beyond-Spec Testing	Pushing all 5 dimensions to "Cliff Effects."

Questions?

Solidifying the 3djax Policy Definition

MDNRef Model Development • Channel Operating Margin Evolution

Image Sources



<https://wirelesspi.com/wp-content/uploads/2023/06/figure-differentiator-impulse-frequency-response-wideband.png>

Source: wirelesspi.com



<https://media.geeksforgeeks.org/wp-content/uploads/20250905101108541893/GMM.webp>

Source: www.geeksforgeeks.org